

Full Length Article

Prediction of lack-of-fusion porosity for powder bed fusion[☆]Ming Tang^a, P. Chris Pistorius^{a,*}, Jack L. Beuth^b^a Department of Materials Science and Engineering, Carnegie Mellon University, 5000 Forbes Avenue, Pittsburgh, PA, 15213, USA^b Department of Mechanical Engineering, Carnegie Mellon University, 5000 Forbes Avenue, Pittsburgh, PA, 15213, USA

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ABSTRACT

A geometry-based simulation is used to predict porosity caused by insufficient overlap of melt pools (lack of fusion) in powder bed fusion. The inputs into the simulation are hatch spacing, layer thickness, and melt-pool cross-sectional area. Melt-pool areas used in the simulations can be obtained from experiments, or estimated with the analytical Rosenthal equation. The necessary material constants, including absorptivity for laser-based melting, have been collated for alloy steels, aluminum alloys and titanium alloys. Comparison with several data sets from the literature shows that the simulations correctly predict process conditions at which lack-of-fusion porosity becomes apparent, as well as the rate at which porosity increases with changes in process conditions such as beam speed, layer thickness and hatch spacing.

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1. Background

Limiting part porosity is essential for building parts with good mechanical properties. The approach described in this paper can be used to predict whether processing conditions in a powder bed fusion process would lead to porosity due to lack of fusion. Other causes of porosity in powder bed fusion include entrapped gas and keyholing; these are not addressed by this prediction method.

Although internal porosity of metal parts manufactured by selective laser melting has been studied for many years, predicting and controlling porosity inside the products still remains a challenge. Numerous studies have been carried out to assess the effect of processing parameters on part density (which is a direct indication of the volume fraction of internal porosity) [1]. Part porosity is often correlated with the process parameter “Energy density”, usually defined as volumetric power input (see Equation (1)) [1–3].

$$\text{Energy density} = \frac{P}{VHL} \quad (1)$$

where P is the beam power, V the beam speed, H hatch spacing and L layer thickness.

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Typical experimentally measured curves of part density (or porosity) with respect to energy density [1] show increased porosity due to keyholing at high energy density (high power and low speed), minimum porosity at intermediate energy density, with the porosity increasing again (due to incomplete melting, or “lack of fusion”) at lower energy density; see Fig. 1.

The fundamental origin of lack-of-fusion porosity is insufficient overlap of successive melt pools. In the approach present here, melt-pool overlap is calculated to predict lack-of-fusion porosity. The fundamental parameters used in the calculation are melt-pool cross-sectional dimensions, hatch spacing and layer thickness. The approach allows rapid prediction of the effects of changes in processing parameters (beam power and beam speed, hatch spacing and layer thickness) on part porosity.

2. Calculation approach

The fundamental principle behind the calculation is that, for complete melting, the depth of the overlap between laterally adjacent melt pools (L^* ; see Fig. 2) must be at least as large as the layer thickness (L) used in building the part. If $L > L^*$, the part will contain lack-of fusion porosity; in that case the volume fraction of lack-of-fusion porosity can be calculated using the approaches described in this paper.

2.1. Analytical relationships

Analytical relationships were derived to guide simulations and processing choices. Each melt pool contains previously deposited

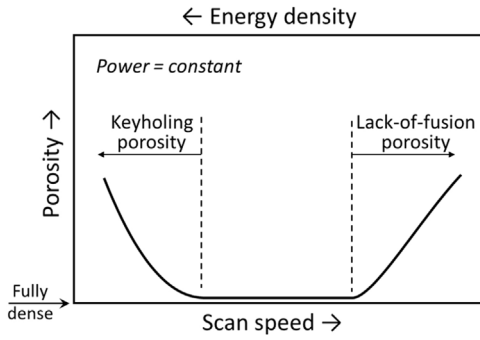


Fig. 1. Typical trend of part porosity with changes in scan speed and energy density (at constant power). Drawn after the results of Gong et al. [1].

material which is remelted in the current beam traverse, and newly deposited material (powder which is melted into the pool). The newly deposited material forms a “cap” (or “reinforcement” in welding terminology [4]) on top of the previously deposited layers. To derive the analytical relationships, the remelted material and the cap were assumed to be half-ellipses, with their horizontal major axes both equal in length to the melt-pool width (W) and located at the surface of the previously deposited layer (see Fig. 2). This geometry was found to approximate many real melt pools, and greatly simplifies analytical treatment. (In the simulation approach described later, any melt-pool cross-section shape can be used in principle, although the simulation examples shown are for the dual half-elliptical melt-pool cross-section depicted in Fig. 2.)

2.1.1. Criterion for full melting

For this dual half-elliptical melt-pool cross-sectional shape, the same relationship previously given for half-elliptical cross-sections [5] states the conditions for which $L^* \geq L$:

$$\left(\frac{H}{W}\right)^2 + \left(\frac{L}{D}\right)^2 \leq 1 \quad (2)$$

where H is the hatch spacing, W the melt-pool width, L the layer thickness, and D the total melt-pool depth. That is, the combinations of hatch spacing and layer thickness which would give full melting (no lack-of fusion porosity) depend only on the melt-pool width and the total melt-pool depth; there is no dependence on cap height (R in Fig. 2) as such.

For process conditions which do not obey Equation (2), lack-of-fusion porosity would be present in the built part. The simulations discussed later in this paper were used to calculate the expected porosity for such conditions.

2.1.2. Build rate, number of melting cycles and cap height

The build rate (part volume deposited per unit time, including porosity) is given by:

$$\text{Buildrate} = HL V \quad (3)$$

The cross-sectional area of a dual half-elliptical melt pool depends on just its width and total depth (and not on the cap height):

$$\text{Area} = \pi DW/4 \quad (4)$$

Because of the overlap between melt pools, material deposited in the part is in general remelted several times. The average number of melting cycles is equal to the rate at which the melt-pool volume propagates through the part ($\text{Area} \times V$) divided by the build rate:

$$\text{Number of melting cycles} = (\pi/4) (W/H) (D/L) \quad (5)$$

For process conditions where the melt pools overlap fully laterally ($H < W$) and in the z direction (that is, conditions that obey Equation (2)), the height of the cap (R) is found from melt-pool geometry, recognizing that the cross-sectional area of material deposited in each bead is given by the shaded region in Fig. 2. This leads to the relationship given in Equation (6):

$$R = \frac{2L}{\sqrt{1 - (H/W)^2} + (W/H) \sin^{-1} \left(\frac{H}{W} \right)} \quad (6)$$

2.2. Simulation principle

A square two-dimensional slice (in the build direction) through a built part was simulated (see Fig. 3). The melt-pool shape was as given in Fig. 2. The calculation inputs were the melt-pool dimensions (W , D and R ; see Fig. 2), hatch spacing (H), layer thickness (L), and the angle between the beam scanning direction of subsequent layers (“hatch rotation”). Consecutive melt pools were placed within the simulation region a layer at a time, as in the actual powder-bed processes. For each layer, melt pools were placed from left to right; the position of the leftmost melt pool was randomized. The apparent width of each melt pool on the cross-section was calculated from the beam direction of that layer (as determined by the number of layers deposited and the hatch rotation). For the results shown here, all melt pools were assumed to have the same width and depth (for one set of building conditions); however, the practically observed variation in melt-pool size along a single bead [6,7] could be readily incorporated in the simulations.

The number of pixels in the simulated area could be adjusted; the simulations shown here were for 1000 pixel side lengths, with

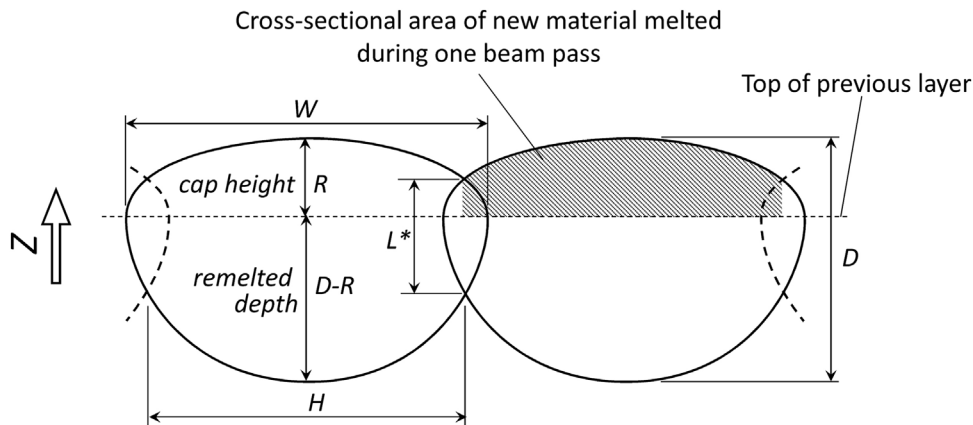


Fig. 2. Schematic of the assumed dual half-ellipse shape of melt pools, showing the overlap between two adjacent beads (for a cross-section in the build direction, perpendicular to the beam scanning direction), and with relevant dimensions labeled.

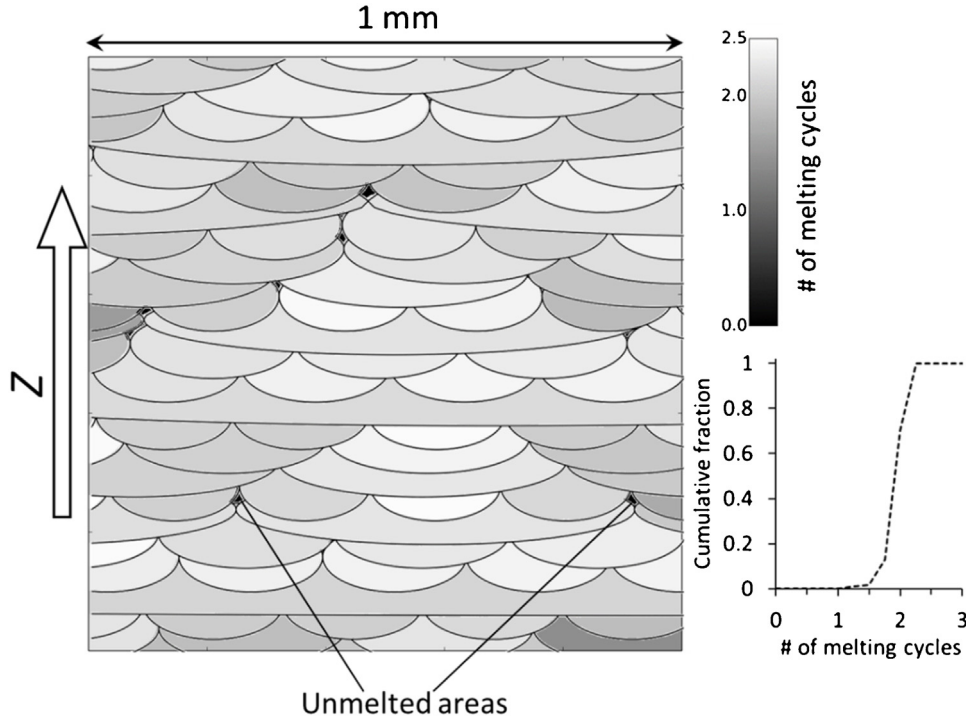


Fig. 3. Example of simulated section. Each melt pool is shaded to indicate the average number of times the material was melted (using the grey scale shown at right.) Melt-pool boundaries drawn in for clarity. The graph at the bottom right gives the distribution of the number of melting cycles for all the melt pools. Assumed dimensions: $W = 220 \mu\text{m}$; $D = 110 \mu\text{m}$; $R = 51 \mu\text{m}$; $H = 220 \mu\text{m}$; $L = 40 \mu\text{m}$; hatch rotation: 67° .

each pixel corresponding (in these examples) to a distance of $1 \mu\text{m}$. For each pixel, the number of times it was within a melt pool was totaled. Each melt pool was assumed to be well-mixed within the plane of the slice, with an average number of melting cycles assigned to all pixels within that melt pool.

The simulation was implemented in Python; the libraries matplotlib and numpy [8] facilitated calculating and displaying the simulation results. Typically at least ten simulations were performed for each set of processing conditions to obtain reliable estimates of the average unmelted volume.

For faster estimates of the porosity, calculations were based on the same dual half-elliptical melt-pool cross-section, but instead calculating the probability that positions in a plane at a particular vertical position would not be melted during deposition of any of the layers. The principle of the calculation is illustrated in Fig. 4. As an example, Fig. 4 illustrates that the probability that positions at the top of layer 1 would remain unmelted after the first pass is simply given by δ_{01}/H , where δ_{01} is the shortest distance between the melt pools at the top of this layer ($\delta_{01}/H = 1$ for the melt-pool geometry shown). This position would have a probability δ_{02}/H of not being melted during deposition of the next layer, and of δ_{03}/H when the third layer was deposited. If the positions of unmelted areas between subsequent layers were independent (that is, with random offsets between melt-pool positions), the probability of this position remaining unmelted after three passes would be given by $(\delta_{01}/H) \times (\delta_{02}/H) \times (\delta_{03}/H)$. Similarly, positions one layer thickness (L) below the top of layer 1 would have a probability $(\delta_{L1}/H) \times (\delta_{L2}/H)$ of remaining unmelted (for the example shown, layer 3 did not extend down to this level). To calculate the total porosity, this calculation was performed for 100 positions distributed vertically over the distance L from the top of layer 1, using numerical integration to find the overall probability of positions in one layer remaining unmelted; the calculation was implemented in a spreadsheet. The distances (δ) were found readily, based on the geometry of the dual half-elliptical melt pools.

The example in Fig. 4 is for the case where the layer thickness is between one-half and one-third the melt-pool depth, but the approach can be readily extended to any value of L/D . The figure was drawn for zero rotation between subsequent layers, but the probability of not being melted by subsequent layers is independent of rotation angle.

The full (Python) and faster (spreadsheet) simulations gave the same estimates of unmelted volume fractions, for the same input values. (This comparison is presented later, in Fig. 10.)

The main result of each simulation was the fraction of unmelted volume in the part, ϵ . The unmelted fraction was calculated as the area fraction of unmelted pixels in the Python simulation, and the probability of remaining unmelted in the spreadsheet calculation. The unmelted fraction was converted to an estimated porosity by assuming the unmelted volume to be filled with powder, and by using an assumed density of fully melted regions. Based on literature results on the porosity in powder beds (0.36 from [7], 0.44 from [9], 0.30 from [10]), the volume fraction of pores in unmelted volumes was taken to be 0.36. The fully melted regions of the part were assumed not to be fully dense, since inspection of literature results indicated that parts typically are approximately 99% dense for process conditions which should eliminate lack-of-fusion porosity. Based on these two considerations, the density of the part relative to fully dense material was calculated as follows:

$$\rho_{\text{part}}/\rho_{\text{true}} = 0.64\epsilon + 0.99(1 - \epsilon) \quad (7)$$

In cases where experimental data on the maximum relative density (other than 99%) were available, these were used instead (examples are listed in Table 3).

2.3. Input parameters: melt-pool geometry

As noted above, the main input parameters were the melt-pool dimensions (width and depth), hatch spacing, layer thickness and hatch rotation. While the latter three parameters are generally con-

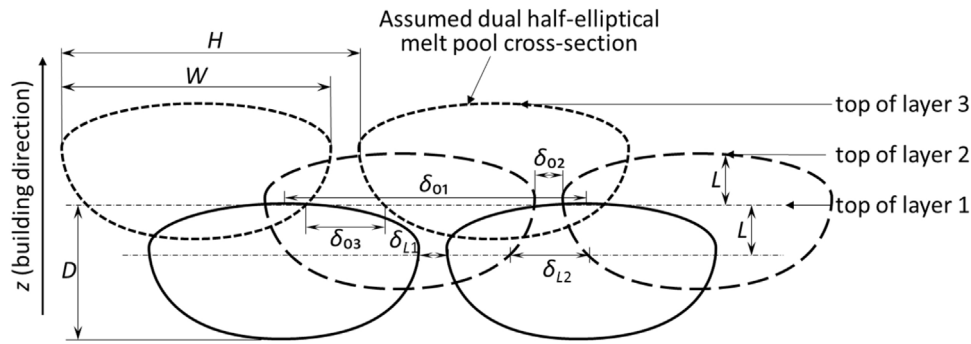


Fig. 4. Geometric relationships used in the simplified approach to estimating the probability of different positions in the part remaining unmelted. (Shown for zero rotation between the scanning directions of consecutive layers.).

Table 1

Alloy properties used in Rosenthal equation [19,20].

Material	Absorptivity ^a (–)	Melting temperature ^b (K)	Thermal conductivity (W/mK)	Density (kg/m ³)	Heat capacity (J/kgK)
AlSi10Mg	0.32	849	110	2670	915
Maraging steel	0.35	1703	15	8000	450
Stainless steel	0.35	1678	14	7800	460
TiAl6V4	0.48	1913	6.7	4430	526

^a Absorptivity here is the average value based on summarized results in Table 2.

^b The melting temperature is the average of the liquidus and solidus temperatures.

trolled directly, the melt-pool dimensions are not. The melt-pool size depends primarily on absorbed power, beam speed and the thermal properties (conductivity and diffusivity) of the material [11]. As with the general approach to process mapping [11], two approaches can be used: the melt-pool dimensions can be measured experimentally, or these can be estimated by calculation. For the examples shown here, melt-pool dimensions were estimated using the simple analytical Rosenthal equation [12].

The Rosenthal equation relies on several restrictive assumptions [12,13]. Notably, only conduction is considered, heat of fusion is not included, thermal properties do not depend on temperature, and the calculation applies to remelting (no addition of powder). However, if the Rosenthal equation is not used to estimate temperatures inside the melt, and only to calculate temperatures in the part up to the boundary of the melt pool, the estimated melt-pool sizes have been found to be realistic and close to experimentally measured values, as shown below.

The applicability of Rosenthal equation was confirmed by the melt-pool shape (semi-circular in cross-section for remelting conditions [5]) and size [5,14,15]. Some examples are shown in Fig. 5. Alloy properties used for this comparison are listed in Table 1. Note that these are room-temperature properties. The Rosenthal equation uses temperature-independent properties; it was found that the properties at room temperature give the best agreement with observed melt-pool sizes for laser melting.

The experimental results used in testing this approach were all for laser melting. The laser wavelength for all the experimental results (from the literature) considered here was 1.06 μm . For these calculations, absorptivity (that is, the fraction of the laser power absorbed by the part) was estimated from literature values, as summarized in Table 2. The absorptivity values are for interaction of the beam with the surface of the melt pool, since for the typical processing conditions the melt pool extends further (in the travel direction) than the beam radius [5]. While the absorptivity is expected to be somewhat affected by other factors such as processing parameters (P and V), melt-pool surface roughness and oxide coverage, and gas composition above the melt pool, the absorptiv-

Table 2

Summary of absorptivity values from literature.

Material	Absorptivity	Remarks	Reference
AlSi10Mg	0.22	Al	King et al. [21]
	0.3	AlSi10Mg	Weingarten et al. [22]
	0.3 ^a	AlSi10Mg	Krishnan et al. [23]
	0.39 ^a	AlSi10Mg	Thijs et al. [14]
	0.39 ^a	AlSi10Mg	Tang et al. [5]
Steel	0.3	304L	Rai et al. [24]
	0.3	316L	Gusarov et al. [25], Masmoudi et al. [9], Kruth et al. [26]
	0.3	solid iron	Khairallah and Anderson [27]
	0.33	316L	Yadroitsev et al. [7]
	0.36	solid iron	King et al. [28]
	0.4	316L	Aboulkhair et al. [29]
	0.4	316L	Aboulkhair et al. [29]
TiAl6V4	0.36	Ti	Verhaeghe et al. [30]
	0.44	TiAl6V4	Aboulkhair et al. [29]
	0.64	Ti	King et al. [21]

^a Absorptivity calculated from the Rosenthal equation using the measured melt-pool width [23]: 156 μm width at 180W laser power and 0.9 m/s velocity; [14]: 170 μm width at 200W laser power and 1.4 m/s velocity; [5]: 278 μm width at 360W laser power and 1.0 m/s velocity).

ity was assumed to be constant for a given alloy for the calculations presented here.

Differences in absorptivity would change the melt-pool size; all else being equal, a higher absorptivity would give a larger melt pool. An analytical expression indicates that the melt-pool width is approximately proportional to the square root of the absorbed power (see the Appendix A for the derivation). This means that the melt-pool width would be approximately proportional to the square root of the absorptivity (all else being equal). Given this relationship, the average absorptivity values of Table 1 would give a good first estimate of melt-pool size unless builds were made under extreme conditions (such as an excessive beam power and

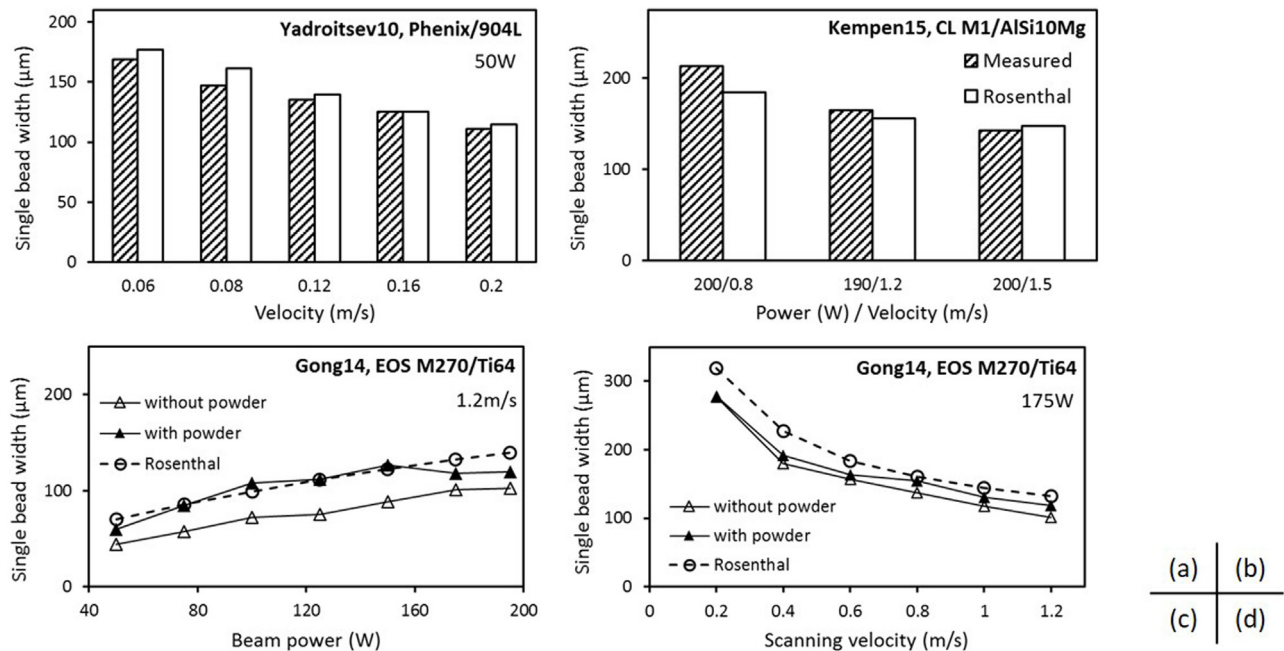


Fig. 5. Comparison of experimentally measured melt-pool widths (single beads) with Rosenthal equation predictions. References are [a-[7],b-[17],c&d-[18]] The data source, SLM machine and powder materials are listed in the top right corner of each figure. Beam power and velocity are the only two variables. Experimental results were measured from the micrographs provided in the paper. Alloy properties used in Rosenthal equation are listed in Table 1.

low speed, which could lead to key-holing and significantly higher absorptivity).

Use of the Rosenthal equation implies the assumption that the addition of powder does not change the melt-pool width and depth (although the cap height would change; see Fig. 2). The limited data available on widths of melt pools for the same material – with and without powder – do indicate that the difference in melt-pool size (with and without powder) is small – see Fig. 5(c) and (d).

While the melt-pool widths calculated with the Rosenthal equation were similar to the reported measurements, melt-pool cross-sectional areas from the Rosenthal equation were generally smaller (by around 30%) than the experimental results. This was especially the case for low energy densities (compared with standard powder-bed fusion conditions) and very high energy densities. Small energy densities imply small melt-pool sizes, which may invalidate the assumption that the incident beam does not interact with powder. Laser-powder interaction is expected to increase the absorptivity. On the other hand, with very large energy densities heat transfer to the metal parts changes from conduction mode to key-holing, in which case the presence of vapor inside the keyhole would increase the effective absorptivity and penetration depth [16]. The effect of a slight larger melt-pool depth was tested by simulation; the results are reported later in this paper.

3. Results and discussion

3.1. Comparison of measured and predicted porosity

In these simulations, it was assumed that only insufficient melt-pool overlap causes lack-of-fusion porosity, and that powder flow is not limiting. To test the simulation model predictions, experimental density data of test pieces from the literature was used for comparison. Three materials (AlSi10Mg, TiAl6V4 and steel) and machines of two manufacturers (EOS and Concept Laser) were considered, with detailed processing parameters and main variables summarized in Table 3. For these comparisons, the melt-pool sizes

were estimated with the Rosenthal equation, using the material properties listed in Table 1.

The simulations used the scan patterns employed when building the parts for which densities were reported. Specifically, hatch spacing, layer thickness and hatch rotation were all considered.

Fig. 6 illustrates the success of this model, with measured densities (solid lines with filled markers) and calculated values (dashed lines with open markers) matching very well. The available experimental data allowed the effects of all processing parameters to be tested, including power (Fig. 6a and c), scanning speed (Fig. 6a–e), hatch spacing (Fig. 6a and d) and layer thickness (Fig. 6d–f). The simulations correctly predicted both the process conditions beyond which porosity started to increase, and rates of porosity increase (with higher beam speed, hatch spacing, or layer thickness).

It should be noted that additional scanning strategies commonly used during fabrication were not included in the present model. First, layer remelting would help to reduce final part porosity, but none of the builds considered here (in Table 3 and Fig. 6) employed remelting during part fabrication. Second, contour scans were not considered in the model, since the contours only account for a small volume fraction of the printed samples (part sizes were at least $10 \times 10 \times 10$ mm in the literature cited); in general, contour scans may be expected to yield larger melt pools and hence locally less lack-of-fusion porosity. Third, island scanning (as employed by the Concept Laser machine) is not expected to have a significant effect on the overall porosity, as the island size was large compared with the part size for the builds considered here ($5 \text{ mm} \times 5 \text{ mm}$ islands for a part with a $20 \text{ mm} \times 20 \text{ mm}$ sample cross-section) [31]; for these conditions, overlap between islands would not give a noticeable difference in overall lack-of fusion porosity.

The effect of such scanning strategies (islands or overlapped stripes) could be included readily in the simulation. Similarly, changes in melt-pool size at the start and end of scans could be simulated, and would be valuable for simulation of complete parts. However, the available experimental data did not allow the effect of such transient changes in melt-pool size to be evaluated. Another

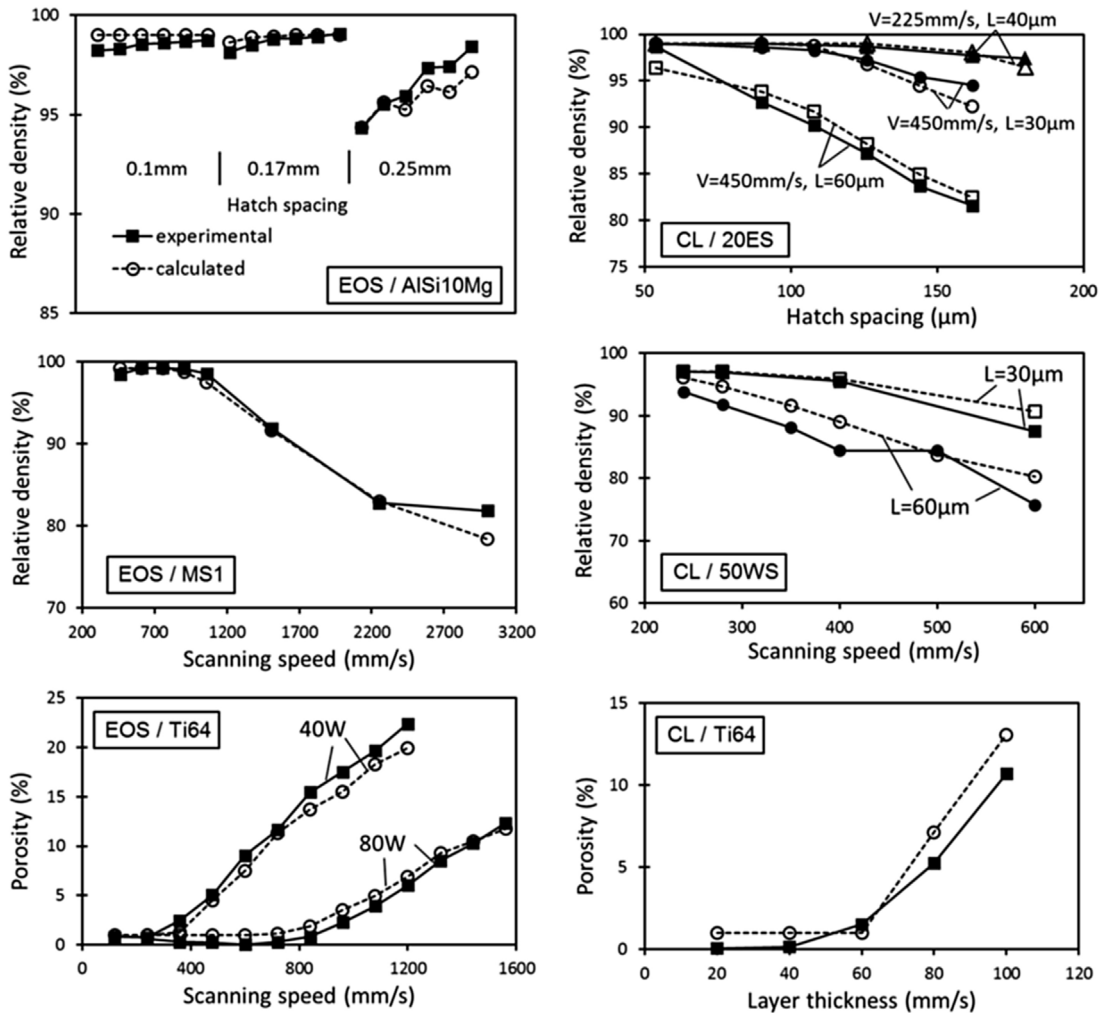
Table 3

Processing parameters from literature data for model validation shown in Fig. 6.

No. in Fig. 6	Reference	Machine	Material	Laser power (W)	Scan speed (mm/s)	Hatch spacing (μm)	Layer thickness (μm)	Max relative density (%)
a	[23]	EOS M270	AlSi10Mg	180, 195	700, 800, 900	100, 170, 250	30	99.1
b	[31]	EOS M270	MS1 ^a	200	400–3000	100	40	99.2
c	[1]	EOS M270	TiAl6V4	40, 80	240–1560	100	30	100
d	[31]	Concept laser M3	20ES ^a	100	225, 450	50–180	30, 40, 60	99.0
e	[31]	Concept laser M3	50WS ^a	100	240–600	125	30, 60	97.1
f	[32],[33]	Concept laser M2	TiAl6V4	400	2400	75	20 ~ 100	99.1

^a MS1 and 50WS are maraging steels and 20ES is a stainless steel.**Table 4**Standard processing parameters for machines and materials, showing melt-pool depth estimated with the Rosenthal equation ($D_{\text{Rosenthal}}$).

Machine	Material	Reference	Laser power (W)	Scan speed (mm/s)	Hatch spacing (μm)	Layer thickness (μm)	$D_{\text{Rosenthal}}$ (μm)
Concept laser M3	50WS (maraging)	[31]	100	200	125	30	83
Concept laser M1	AlSi10Mg	[14]	200	1400	105	30	75 ^a
EOS M270	MS1 (maraging)	[31]	200	750	100	40	65
EOS M270	TiAl6V4	[34]	170	1250	100	30	64
EOS M270	GP1 (stainless)	[2]	195	800	100	40	63
EOS M270	AlSi10Mg	[23,34]	195	800	170	30	90
EOS M280	AlSi10Mg	[13]	370	1300	190	30	113 ^a

^a Recalculated values using an absorptivity of 0.32, as noted in Table 1.**Fig. 6.** Comparison of measured (literature data) and predicted (this work) part density and porosity. Solid lines and filled markers are experimental results, and dashed lines with open markers are simulation results (this work). See Table 3 for references and process conditions.

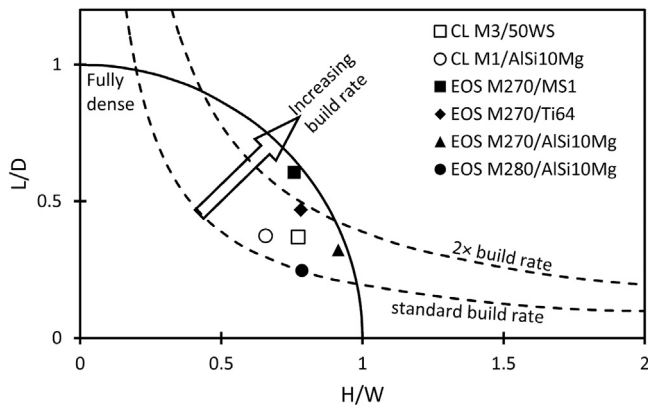


Fig. 7. Processing map of hatch spacing and layer thickness (relative to melt-pool dimensions); conditions within the circular arc would avoid lack-of-fusion porosity. Markers indicate standard building conditions for different machine/material combinations (see Table 4 for references).

useful addition to the model would be to consider local part geometry (such as overhangs or thin walls) which would change heat transfer conditions, and hence give different melt-pool sizes. For such cases, the prediction method would remain the same (based on melt-pool overlap), and would consider the particular scanning pattern and estimated melt-pool sizes for the particular heat transfer conditions.

3.2. Processing map

Equation (2) describes a simple relationship between values (L/D) and (H/W) which would avoid lack-of fusion porosity; graphically, the relationship is in the form of a circular arc (if L/D is plotted vs. H/W); see Fig. 7. For comparison with this prediction, Table 4 summarizes the standard processing parameters for producing parts of various materials, which conditions are also included in the map of Fig. 7. For plotting these, the melt-pool depth was estimated with the Rosenthal equation with absorptivity and alloy properties as in Table 1. If Fig. 7 is drawn for constant values of power and speed, the broken lines are contours of constant build rate, and also constant energy density. The contour labeled “standard build rate” refers to the processing parameters for EOS M280/AlSi10Mg, as studied in this research [13]. Fig. 7 demonstrates that most standard processing conditions are well within the safe range for producing fully dense products (with in many cases similar H/W values of around 0.74).

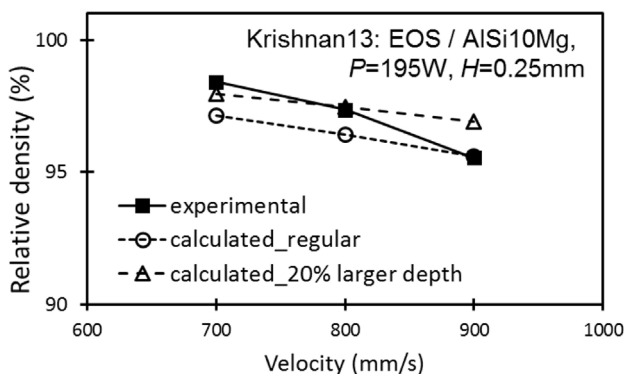


Fig. 9. Effect of a 20% increase in melt-pool depth (without changing the total melt-pool width) on the predicted part density, compared with measurements [23]; conditions are for case (a) in Fig. 6.

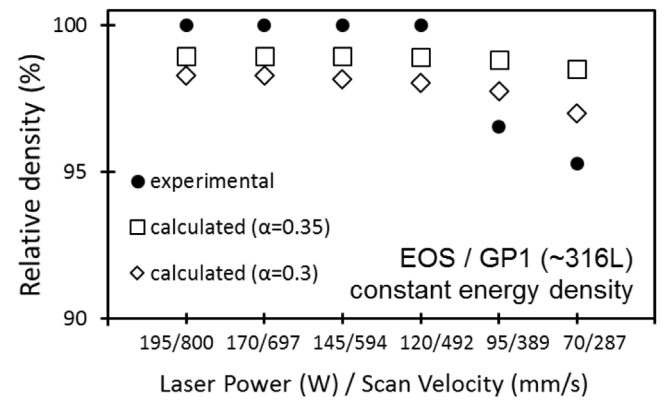


Fig. 8. Densities of GP1 (stainless steel) parts built at the same energy density but different combinations of laser power and speed, showing experimental measurements [2] and simulation results (this work). In the legend, α is laser absorptivity.

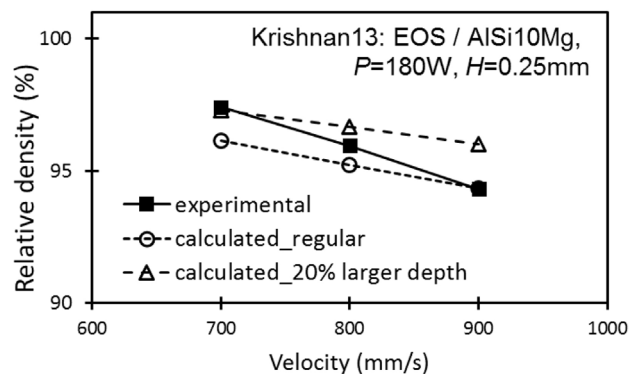
One implication of where standard conditions plot in Fig. 7 is that in many cases there is room to increase the build rate without causing lack-of-fusion porosity.

3.3. Porosity changes at the same energy density

As mentioned above, the broken lines in Fig. 7 are contours of constant building rate and of constant energy density (if the processing map is drawn for constant power and speed). The energy-density contours intersect the circular arc which defines the minimum condition for parts with no lack-of-fusion porosity. The implication is that increases in energy density need not necessarily improve part density, and that maintaining a constant energy density (while changing other processing conditions) may result in varying porosity. Literature results for parts built at the same energy density but different power/velocity combinations [2] were used to test this prediction, with the results given in Fig. 8. Both the measurements and the simulations demonstrate that maintaining the same energy density does not guarantee a constant porosity. (Simulation results are shown for two values of absorptivity – 0.3 and 0.35 – because of the wide range of laser power in this data set; in general, it is expected that the absorptivity would be somewhat smaller at lower laser power [35].)

3.4. Effect of varying melt-pool depth

As mentioned earlier, the Rosenthal calculations often underestimated the melt-pool depth. To test the effect of this underestimate, simulations were repeated with 20% larger melt-pool depths for one machine/material combination at two values



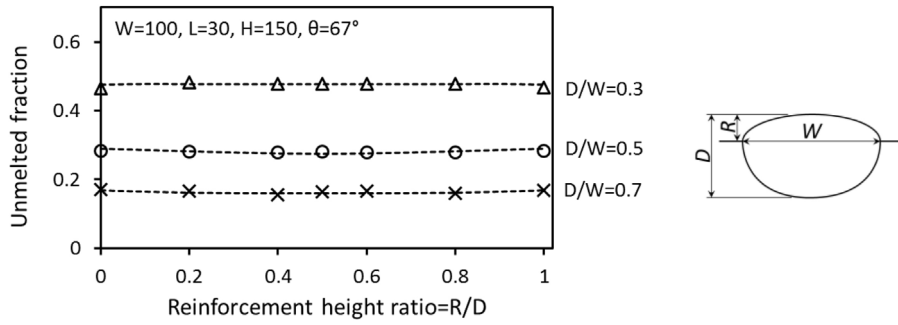


Fig. 10. Predicted unmelted volume fraction for various ratios of cap height to melt-pool depth, different ratios of melt-pool depth to width, and hatch rotation angle of 67° . Broken lines are the results of the faster (spreadsheet) simulation, and the markers are the results of the full (Python) simulation. The melt-pool width was constant at 100 (arbitrary length units).

of laser power and different speeds. Fig. 9 shows the expected lower porosity (larger density) with increased melt-pool depth, although the effect is not large. Certainly, improved estimates or measurements of melt-pool depth and width would improve the accuracy of the predictions of this geometry-based simulation.

3.5. Effect of cap height

The criterion for full melting (Fig. 3) does not depend on the cap height. However, for conditions where this criterion is not met (that is, when lack-of-fusion porosity would be present), it might be expected that the detail of the melt-pool shape would affect the total porosity. To test this, full simulations were performed for different cap heights, and for melt pools with different depth-to-width ratios. These results were compared with the faster (spreadsheet) calculations (Fig. 4) for the same melt-pool geometries. In Fig. 10 the results of the faster calculation are shown by broken lines and those of the full simulation with markers; the predictions of the two simulations agree. The simulations also indicate that there is no significant effect of cap height on the unmelted volume fraction (although the shapes of unmelted volumes would vary).

4. Conclusion

A geometry-based simulation model has been developed to predict the volume fraction of unmelted material in parts built by power bed fusion. The input data required by the model are the shape of the melt-pool cross-section, hatch spacing and layer thickness. Comparison with several sets of experimental data from the literature shows that the model correctly predicts process conditions beyond which lack-of-fusion porosity would be present, and the numerical value of porosity if melting is incomplete.

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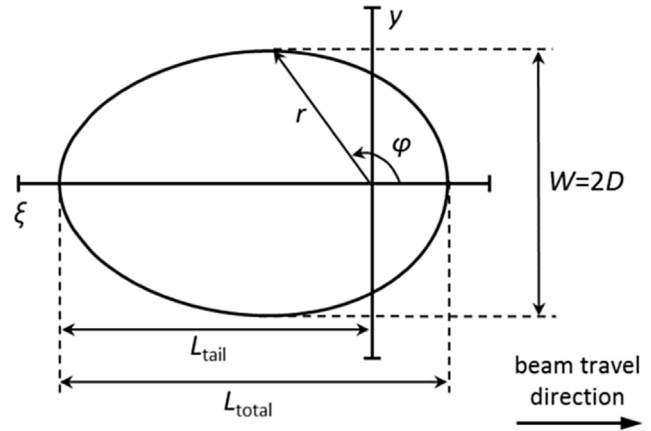


Fig. 11. Schematic of melt-pool dimensions (viewed from above; frame of reference moves with the beam). The beam is focused at the intersection of the y and ξ axes.

Appendix A. Approximate analytical expression for melt-pool width

The Rosenthal equation [12] gives the shape of the melt pool (shown in Fig. 11)

$$T_f = T_0 + \frac{Q}{2\pi kr} \exp \left[-\frac{V(\xi + r)}{2\alpha} \right] \quad (8)$$

where T_f is the melting point of the alloy, T_0 the temperature far from the melt pool, Q is the absorbed power, k the thermal conductivity of the alloy, V the beam speed, and α is the thermal diffusivity of the alloy. The dimensions r and ξ define the size and shape of melt pool, where ξ is the distance from the heat source, measured in the direction of beam travel, and $r = (\xi^2 + y^2)^{0.5}$ is the distance to the edge of the melt pool from the heat source. At the position of maximum melt-pool width, $r = r^*$.

To simplify the expressions, the auxiliary constants M and N are defined as shown in the equations below. Both constants are positive and have the dimension $1/\text{length}$. For typical powder bed fusion conditions and alloys, M is much larger than N , with AlSi10Mg as an exception; see Table 5.

(M/N is smaller for AlSi10Mg because of the high thermal conductivity of this alloy.)

$$M = \frac{V}{2\alpha} \quad (9)$$

$$N = \frac{2\pi k(T_f - T_0)}{Q} \quad (10)$$

Table 5

Typical values of parameters in the analytical expression for melt-pool width: r^* is the distance from the heat source to the melt-pool edge at the position of maximum width (values found by numerical solution of the Rosenthal equation for typical powder bed fusion conditions, taken to be $Q=100$ W, $V=1$ m/s, $T_0=303$ K); room-temperature alloy properties were used to calculate $M=V/2\alpha$ and $N=2\pi k(T_f-T_0)/Q$.

	AlSi10Mg	Maraging steel	Stainless steel	Ti64
r^* (μm)	138	267	269	548
M/N (–)	2.9	91	106	257
r^*M (–)	1.5	32	34	95
$[-\ln(r^*N)]/Mr^*$	0.42	0.03	0.03	0.01

Using these constants, the Rosenthal equation can be rewritten as follows:

$$\ln Nr + M(\xi + r) = 0 \quad (11)$$

To find the melt-pool width, it is convenient to express the equation in cylindrical coordinates, with $\xi = r \cos \varphi$ and $y = r \sin \varphi$; substitution gives the following expression:

$$\ln Nr + Mr(\cos \varphi + 1) = 0 \quad (12)$$

The melt-pool width is twice the value of y where $dy/d\xi = 0$; in terms of the cylindrical coordinates this yields the following expression:

$$\begin{aligned} \frac{dy}{d\xi} &= \frac{dr \sin \varphi + r d\varphi \cos \varphi}{dr \cos \varphi - r d\varphi \sin \varphi} = 0 \\ \Rightarrow \frac{dr}{d\varphi} &= -\frac{\cos \varphi}{\sin \varphi} r \end{aligned} \quad (13)$$

Differentiation of the Rosenthal equation (Equation (12)) gives another expression for $dr/d\varphi$:

$$\frac{dr}{d\varphi} = \frac{r \sin \varphi}{\frac{1}{Mr} + \cos \varphi + 1} \quad (14)$$

Equating these two expressions gives the relationship between r and φ at the melt-pool width; the specific values of r and φ at this position are denoted with asterisks:

$$\cos \varphi^* = \frac{-r^*M}{1 + r^*M} \quad (15)$$

Substitution in the Rosenthal equation then yields

$$\begin{aligned} \ln r^*N + r^*M(1 + \ln r^*N) &= 0 \\ \Rightarrow 1 + \ln r^*N &= -\frac{\ln r^*N}{r^*M} \end{aligned} \quad (16)$$

For typical powder bed fusion conditions, for all alloys except AlSi10Mg, $-(\ln r^*N)/r^*M \approx 0$ (see Table 5), which means that r^* is approximated by:

$$r^* \approx \frac{1}{eN} \quad (17)$$

The half-width ($W/2$) of the melt pool is given by the value of y where $r=r^*$ and $\varphi=\varphi^*$, that is

$$W/2 = r^* \sin \varphi^* \quad (18)$$

From the relationship between r^* and φ^* , the half-width is then given by:

$$\begin{aligned} \sin \varphi^* &= \frac{\sqrt{1}}{1 + r^*M}; \\ W/2 &= r^* \sin \varphi^* = r^* \frac{\sqrt{1}}{1 + r^*M} \end{aligned} \quad (19)$$

Because $r^*M \gg 1$ for all alloys except AlSi10Mg, this can be approximated as follows:

$$W/2 \approx \sqrt{\frac{2r^*}{M}} \approx \sqrt{\frac{2}{eMN}} = \sqrt{\frac{2Q}{e\pi\rho C(T_f - T_0)V}} \quad (20)$$

This expression can be rewritten in terms of the power required to achieve a melt-pool of width W , as follows:

$$Q \approx e\pi\rho C(T_f - T_0)WV^2/8 \quad (21)$$

For the limiting case where the heat source is stationary, the Rosenthal equation can be used to find the power required to maintain a melt pool of width $W=2r^*$, as follows:

$$Q_{V=0} = 2\pi k(T_f - T_0)r^* = \pi k(T_f - T_0)W \quad (22)$$

It was found that the approximation is improved by adding this zero-speed power to the power for non-zero beam speed, as follows:

$$Q \approx \pi k(T_f - T_0)W + e\pi\rho C(T_f - T_0)WV^2/8 \quad (23)$$

(for alloys other than AlSi10Mg)

It was found that this expression can also be used to approximate the melt-pool widths for AlSi10Mg, if the second term is multiplied by a factor of 0.8:

$$Q \approx \pi k(T_f - T_0)W + e\pi\rho C(T_f - T_0)WV^2/10 \quad (24)$$

(for AlSi10Mg)

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